

DIVYESH MISTRY

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EDUCATION

Master of Science in Computer Science - Data Science

University of Southern California | 3.67 GPA

Aug 2024 - May 2026

Los Angeles, California

Bachelor of Engineering in Computer Engineering (Minors in Data Science)

University of Mumbai | 9.00 GPA

Aug 2021 - June 2024

Mumbai, India

SKILLS

Languages: Python, Cython, R, Java, Scala, Bash, C++, C#, JavaScript, TypeScript, Dart.

Cloud & Infrastructure: AWS, GCP, Azure, Parallel Processing, Vector Databases.

Frameworks & Visualization: Flask, Streamlit, React Native, Flutter, Django, Tableau, Power BI, Git.

Data Engineering: Apache Spark, ETL Pipelines, Data Mapping/Migration, NoSQL (Cassandra, MongoDB, Firebase)

AI/ML & Data Science: NLP, LLM Fine-Tuning (LoRA/PEFT), Quantization, PyTorch, Data Mining, Statistical Modeling.

WORK EXPERIENCE

ServiceLink

June 2025 - Aug 2025

Data Science Intern

Irving, Texas

- Optimized post-close document analysis accuracy by **38%** by fine-tuning **Gemma-3-4b-it** via **LoRA** and **PEFT** on a custom **4,000+ image dataset**.
- Achieved optimal model architecture selection for **GPU-accelerated inference** by benchmarking 10+ vision-language models, evaluating accuracy, **memory bandwidth**, inference speed and model size.
- Reduced model footprint by **70%** via **4-bit quantization** (INT4), optimizing kernels for resource-constrained **GPU environments** using **LLaMA.cpp**.
- Delivered **real-time**, on-device document analysis across 3+ platforms using a **React Native** app integrated with **llama.rn** for 100% data privacy.

USC Ming Hsieh Department of Electrical and Computer Engineering

Dec 2024 - May 2025

Software Engineer

Los Angeles, California

- Analyzed and resolved training and **inference performance bottlenecks** in a Python-based simulation, re-engineering core modules with **Cython** to achieve a **25% speedup** on high-performance compute nodes
- Engineered the forecasting architecture to support **parallel processing**, enabling the efficient, simultaneous execution of projections for multiple states/regions.

USC Keck School of Medicine

Sept 2024 - Dec 2024

Research Assistant

Los Angeles, California

- Identified **Structural Variants** in Human DNA, aligned oncogene sequences with **whole human genome** using **Sniffles2**, enhancing understanding of genetic variations.
- Translated gene sequences to Amino acid sequences and annotated mutations to identify types with reference tables.

OziBook

Aug 2023 - Oct 2023

Data Analyst Intern

Mumbai, Maharashtra

- Spearheaded data workflows for a 30-member cross-functional team, building a centralized client database that successfully converted 2 new enterprise clients.
- Analyzed **900+ potential client profiles** using **Pandas** and **Tableau** to identify and shortlist high-conversion targets aligned with company services.
- Translated complex datasets into **actionable business stories** using **Tableau** and **Power BI**.

PROJECTS

AI-Powered Academic Research Assistant | [Link](#)

- Built a scalable **NLP framework** to transform **unstructured academic text into structured summaries**, utilizing **Parallel Processing** to reduce data ingestion and summarization latency by **90%**.
- Fine-tuned **LongT5 Transformer model** for summarization using a dataset of **1,000** computer science research papers.
- Integrated a hybrid summarization approach by applying LexRank algorithm to identify Key Takeaways, effectively reducing manual literature review time by 90% through automated extractive-abstractive filtering

Choose Your Own Adventure Game | [Link](#)

- Engineered real-time data injection pipeline into **GPT-3** on **Cassandra Astra**, enabling dynamic, personalized gameplay through contextual AI-driven responses to player choices.
- Optimized game performance by utilizing **vector databases** for efficient data storage and retrieval.
- Leveraged Cassandra chat message history and conversation buffer memory for efficient data handling.
- Showcased prompt templates to dynamically inject variables into AI prompts, fostering customized interactions.

CERTIFICATIONS

- [Data Visualization and Communication with Tableau](#) by Duke University | Coursera **2023**
- [Data Analysis with R Programming](#) by Google | Coursera **2023**
- [React Native - The Practical Guide](#) by Academind | Udemy **2023**
- [Flutter & Dart - The Complete Guide](#) by Academind | Udemy **2022**